Unit: 6 Unit 4: Absolute Value **Name:** ____________

Topic: Absolute Value Function and its Graph **Date:** __________

Challenge (Habanero)

Q1. What is the vertex of the graph of $y = |x - 2| + 1$?
(A) $(2, 1)$
(B) $(1, 2)$
(C) $(0, 1)$
(D) $(-2, 1)$
(E) $(2, -1)$

Q2. The graph of $y = |x + 3|$ has an axis of symmetry at $x =$
(A) -3
(B) 3
(C) 0
(D) -2
(E) 2

Q3. A gaming company has a data usage function $y = |x - 5|cdot2$ where y is the amount of data used and x is the number of hours played. If $x = 3$, then $y =$
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Q4. The function $y = 2|x| - 1$ has a minimum value of
(A) -1
(B) 1
(C) 2
(D) 3
(E) 4

Q5. A coding logic function is given by $y = |x - 2| + |x + 2|$. The value of y when $x = 0$ is
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q6. Which of the following functions has the same graph as $y = |x|$ but shifted 2 units to the right?
(A) $y = |x - 2|$
(B) $y = |x + 2|$
(C) $y = x - 2$
(D) $y = x + 2$
(E) $y = -x + 2$

Q7. A scoreboard function is given by $y = |x - 5| - 3$. The value of y when $x = 8$ is
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q8. The function $y = 3|x| - 2$ has an axis of symmetry at $x =$
(A) -2
(B) -1
(C) 0
(D) 1
(E) 2

Open Ended Questions (FRQ)

Q9. Graph the function $y = |x + 1| - 2$ and identify its vertex, axis of symmetry, and direction of opening. Provide your answer in 5 lines or less.

Q10. A gaming company has a function $y = |x - 3|cdot4$ where y is the number of points scored and x is the number of levels completed. Graph the function and determine the number of points scored when $x = 5$. Provide your answer in 5 lines or less.

Chef's Tips

- Tip 1: Review the properties of absolute value functions and their graphs.
- Tip 2: Use technology to verify student answers.
- Tip 3: Provide opportunities for students to create their own absolute value functions and graph them.